Analysis 1 HW 6 Part 2

6.2

f is continuous on $[a, b]$.
$f \geq 0$.
$\int_a^b f(x)\, dx = 0$.

To prove: $f(x) = 0$ for all $x \in [a, b]$.

Proof. FTSOC, assume $f(p) = q > 0$ for some $p \in [a, b]$. Since f is continuous, there exists $\delta > 0$ such that $|x - p| < \delta \implies |f(x) - q| < q/2$. Consider a partition P with $x_{k-1} = p - \delta/2$, $x_k = p + \delta/2$ for some k.

$$\int_a^b f\, dx \geq L(P, f) \geq m_k \Delta x_k \geq \frac{q}{2}\delta$$

This makes it impossible for $\int_a^b f\, dx$ to be 0. Thus, $f(x) = 0$. □

6.5

$f : [a, b] \to \mathbb{R}$.
f is bounded.

If $f^2 \in \mathscr{R}$ on $[a, b]$, does it follow that $f \in \mathscr{R}$ on $[a, b]$?

No. Consider the function which is defined to be 1 at rational inputs and -1 at irrational inputs. Clearly, $f^2 \in \mathscr{R}$ since it is a constant function. However, $f \notin \mathscr{R}$ since $U(P, f) - L(P, f) = 2(b - a)$ for all partitions P.

If $f^3 \in \mathscr{R}$ on $[a, b]$, does it follow that $f \in \mathscr{R}$ on $[a, b]$?

Yes. $\sqrt[3]{x}$ is a continuous function on any interval in $\mathbb{R}$. It follows from Theorem 6.11 that $\sqrt[3]{f^3(x)} = f(x) \in \mathscr{R}$ on $[a, b]$.

6.8

$f \in \mathscr{R}$ on $[a, b]$ for every $b > a$ where a is fixed.
$\int_a^\infty f(x)\,dx \equiv \lim_{b\to\infty} \int_a^b f(x)\,dx$, if the limit exists in $\mathbb{R}$.

To prove: If $f \geq 0$ and is monotone decreasing on $[1, \infty)$,

$$\int_1^\infty f(x)\,dx \text{ converges } \iff \sum_{n=1}^\infty f(n) \text{ converges.}$$

Proof. Consider the sequence (t_n) defined by

$$t_n \equiv \int_1^n f(x)\,dx.$$

For any n, consider the partition $P = \{1, 2, \ldots, n\}$. We know that $L(P, f) \leq t_n \leq U(P, f)$. Since f is monotone decreasing, we can write

$$\sum_{k=2}^n f(k) \leq t_n \leq \sum_{k=1}^{n-1} f(k).$$

Note that all terms are positive. Now, if $\sum f(n)$ converges, it follows from the comparison test that t_n converges. Say it converges to t. For every $\epsilon > 0$, we can find N such that $n \geq N \implies t - t_n < \epsilon$. Note that $\int_a^x f(x)\,dx$ is a monotone increasing function, since $f \geq 0$. Hence, for $x > N$, $t - \epsilon < \int_a^x f(x)\,dx < t + \epsilon$. Thus, $\lim_{x\to\infty} \int_a^x f(x)\,dx$ exists.

On the other hand, if $\lim_{x\to\infty} \int_a^x f(x)\,dx$ exists, then t_n converges from the sequential criterion for limits, and it follows from the comparison test that $\sum f(n)$ converges. $\square$